

Air University



Organization & Assembly (Lab)

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Lab Task # 09

Question

IF AL <= BL

THEN

DISPLAY THE CHARACTER IN AL

ELSE

DISPLAY THE CHARACTER IN BL

END_IF

Suppose AL and BL contain extended ASCII characters. Display the one that comes first in the character sequence. Write an assembly language program that uses CMP and JNBE instructions to accomplish this task.

Solution

Code

```
.MODEL SMALL
.STACK 100H

; Two extended ASCII chars: C9H (201 'Œ') and CDH (205 '=')

.CODE
MAIN PROC

    ; Load chars into AL and BL
    MOV AL, 201
    MOV BL, 205

    ; Compare AL and BL
    CMP AL, BL

    ; If AL > BL, jump to display_B
    JNBE display_B

    ; Display AL if AL <= BL
display_A:
    MOV DL, AL
    JMP end_if

    ; Display BL if AL > BL
display_B:
    MOV DL, BL

end_if:
    ; Display char in DL
```

```
        MOV     AH, 02H
        INT     21H

; Exit program
        MOV     AH, 4CH
        INT     21H

MAIN    ENDP
END MAIN
```

Explanation

- 1. Loading and Comparison:** The code loads two character codes (201 and 205) into registers AL and BL. The CMP AL, BL instruction performs the subtraction AL - BL internally, setting the CPU flags based on the result.
- 2. Conditional Branching:** The key instruction is JNBE (Jump if Not Below or Equal). Since the condition we want to execute *first* is IF $AL \leq BL$ (display AL), we check the opposite: IF $AL > BL$ (display BL).
 - If $AL > BL$, JNBE is true, and execution jumps to the ELSE block at display_B.
 - If $AL \leq BL$, JNBE is false, and execution proceeds directly to the THEN block at display_A.
- 3. Output:** In either block, the character determined to be first (smaller value) is moved to DL. A JMP ensures the THEN block bypasses the ELSE block. The character in DL is then displayed using the DOS Character Output function (INT 21h, AH=02h).
- 4. Result:** Since 201 is less than 205, the code displays the extended ASCII character corresponding to the code 201 ('⌘').

Output

```
D:\>TLINK D:\test
Turbo Link  Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International

D:\>D:\test
⌘ ←
D:\>{[
```